

Classical Controls

$$t_{r1} \approx 2.2\tau \quad t_{r2} \approx \frac{1.8}{\tau} \quad t_{s1} \approx 4\tau \quad t_{s2} \approx \frac{4}{\zeta\omega_n}$$

$$t_P = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad M_P = \exp\left(\frac{-\pi\zeta}{\sqrt{1 - \zeta^2}}\right)$$

$$\%O.S. = M_P \times 100\% \quad \zeta = \frac{-\ln(\%O.S./100)}{\sqrt{\pi^2 + \ln^2(\%O.S./100)}}$$

$$\delta \approx \ln\left(\frac{y(t)}{y(t + \Delta t)}\right) \quad (\text{Log decrement}) \quad \zeta \approx \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

$$S_P^F = \lim_{\Delta P \rightarrow 0} \frac{\text{fractional change in function } F}{\text{fractional change in parameter } P}$$

$$S_P^F = \frac{P}{F} \frac{\partial F}{\partial P} \quad 0 \equiv \text{not sensitive} \quad 1 \equiv \text{fully sensitive}$$

Op-Amps

$$V_R(s) = RI(s) = Z_R I(s) \quad V_C(s) = \frac{1}{sC} I(s) = Z_C(s) I(s)$$

$$V_L(s) = LsI(s) = Z_L(s) I(s)$$

Take **either** path, not both

1. In general: $V_O = A(V^+ - V^-)$
2. In negative feedback:
 - (a) $V^+ = V^-$
 - (b) No current flows between "V⁺" and "V⁻" terminals

Laplace Transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt$$

$$\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - \sum_{i=1}^n s^{n-i} f^{(i-1)}(0)$$

Theorems

1. Initial value: $\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$
2. Final value: **iff** all roots of $sF(s)$ are negative real, then $f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$

Standard Forms

$$\frac{Y(s)}{R(s)} = K \frac{1/\tau}{s + 1/\tau} \quad \frac{Y(s)}{R(s)} = K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$D(s) : s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \implies s_{1,2} = \zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

Signal Flow Diagram

1. Label nodes (input/output signals, summing junctions)
2. Assign each branch accordingly
3. Label and define forward paths and loops

Mason's Rule

$$\frac{\text{output signal}}{\text{input signal}} = G_x(s) = \frac{1}{\Delta} \sum_i F_i \Delta_i$$

F_i = path gain of i^{th} forward path from input to output

Δ = system determinant

$$\begin{aligned} &= 1 - \sum (\text{all individual loop gains}) \\ &+ \sum (\text{gain products of all possible combinations of **two** loops that do not touch}) \\ &- \sum (\text{gain products of all possible combinations of **three** loops that do not touch}) \\ &+ \sum (\text{gain ... **four** loops ... do not touch}) \\ &- \sum (\text{gain ... **five** loops ... do not touch}) \\ &+ \dots \dots \dots \rightarrow \infty \text{ loops} \\ \Delta_i &= i^{th} \text{ forward path determinant} \\ &= \text{value of } \Delta \text{ for that part of the diagram that does not touch the } i^{th} \text{ forward path } F_i \end{aligned}$$

Routh-Hurwitz Criterion

1. **Fact 1**: If there is at least one sign change in the coefficients of the **monic** polynomial, then there will be at least one pole **not** in the OLHP.
2. **Fact 2**: If all coefficients of **monic** polynomial are positive, then we cannot say anything about the location of the poles.

if $D(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n$, then the RH table will populate like this:

s^n	1	a_2	a_4	$\dots$
s^{n-1}	a_1	a_3	a_5	$\dots$
s^{n-2}	b_1	b_2	b_3	$\dots$
$\vdots$	$\vdots$	c_1	c_2	$\dots$
s^0	$\vdots$	$\vdots$	$\vdots$	$\ddots$

where the b_n and c_n coefficients are:

$$b_1 = -\frac{\begin{vmatrix} 1 & a_2 \\ a_1 & a_3 \end{vmatrix}}{a_1} \quad b_2 = -\frac{\begin{vmatrix} 1 & a_4 \\ a_1 & a_5 \end{vmatrix}}{a_1} \quad b_3 = -\frac{\begin{vmatrix} 1 & a_6 \\ a_1 & a_7 \end{vmatrix}}{a_1}$$

$$c_1 = -\frac{\begin{vmatrix} a_1 & a_3 \\ b_1 & b_2 \end{vmatrix}}{b_1} \quad c_2 = -\frac{\begin{vmatrix} a_1 & a_5 \\ b_1 & b_3 \end{vmatrix}}{b_1} \quad c_3 = -\frac{\begin{vmatrix} a_1 & a_7 \\ b_1 & b_5 \end{vmatrix}}{b_1}$$

sign changes in col. 1 = # poles in ORHP

Special Cases

1. Replace any col. 1 zeroes with ϵ (a teeny number)
2. Replace any row of zeroes with d/ds (auxiliary polynomial) (generated from above row)

Root Locus

RL techniques can be used for any arbitrary parameter Γ , provided the char. eq. is of the form $1 + \Gamma \square = 0$, where $\square$ is adapted to contain any term to make the form fit.

$$1 + K\hat{G}(s) = 0 \quad |K\hat{G}(s)| = |-1|$$

$$\angle(K\hat{G}(s)) = 180^\circ + 360^\circ l, \quad l = 0 \pm 1, \pm 2, \dots$$

$$\begin{aligned} \angle \hat{G}(s) &= \sum \psi_i - \sum \phi_j \\ &= + \sum (\text{angles from zeros to test point}) \\ &\quad - \sum (\text{angles from poles to test point}) \end{aligned}$$

$$m \triangleq \text{number of zeroes} \quad n \triangleq \text{number of poles}$$

1. Left of odd numbered poles/holes on real axis
2. m poles go to m finite zeros, n poles go to $m - n$ infinite zeros

$$\lim_{K \rightarrow \infty} \hat{G}(s) = \lim_{K \rightarrow \infty} -\frac{1}{K} = 0$$

3. The poles move towards the zeros along asymptotes

$$\phi_h = \frac{180^\circ + 360^\circ(h-1)}{n-m} \quad h = 1, 2, \dots, n-m$$

4. The location the asymptotes meet (center)

$$\alpha = \frac{\sum \text{Re}(P_i) - \sum \text{Re}(z_j)}{n-m}$$

5. Angles

$q \triangleq$ order of hole/pole, $q > 1 \Rightarrow$ repeated holes/poles

$$h \triangleq 1, 2, 3, \dots, q$$

Departures

$$q\phi_{dep} = (\sum \psi_i - \sum \phi_j) - 180^\circ - 360^\circ(h-1)$$

ψ_i = angles from zeroes to pole of interest

ϕ_j = angles poles to pole of interest

Arrivals

$$q\phi_{arr} = (\sum \phi_h - \sum \psi_k) + 180^\circ + 360^\circ(h-1)$$

ϕ_{hi} = angles from poles to zero of interest

ψ_k = angles from zeroes to zero of interest

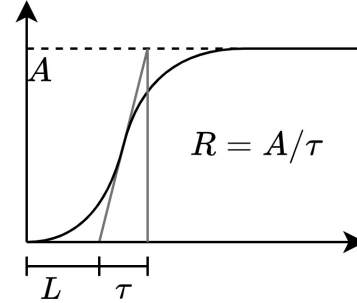
6. The instability point is the K value that causes poles to land on the $j\omega$ -axis. This is found by solving for the K that causes a row of zeroes to appear on a RH table. Then the auxiliary polynomial is solved for the ω at that K value
7. Breakaway/in points are found by manipulating the RL equation to solve for $K(s)$. The s that satisfies $\partial K(s)/\partial s = 0$ is the point

Bode Plots

Ziegler-Nichols Tuning Methods

1. 25% Decay Approach

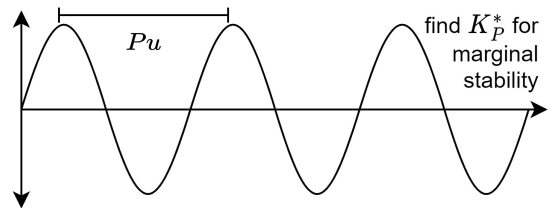
Assume an open-loop system has the following step-response. Finding L, R and selecting the following gains results in a Closed-Loop controller that reduces the oscillating overshoot to 25% over the course of one period, with $C(s) = K_P(1 + 1/T_i s + T_d s)$



	Optimal Gains
P	$K_P = 1/RL$
PI	$K_P = 0.9/RL \quad T_i = L/0.3$
PID	$K_P = 1.2/RL \quad T_i = 2L \quad T_d = L/2$

2. Ultimate Sensitivity Method

Starting with a closed-loop system, with reference = 0, increase K_P until marginal stability and measure Pu and K_P^* . The optimal gains are given for $C(s) = K_P(1 + 1/T_i s + T_d s)$



	Optimal Gains
P	$K_P = K_P^*/2$
PI	$K_P = 0.45K_P^* \quad T_i = Pu/1.2$
PID	$K_P = 0.6K_P^* \quad T_i = Pu/2 \quad T_d = Pu/8$

Bode Plots

Separate the TF into it's constitutive terms, and draw each term's Bode plot based upon it's type. The total Bode plot is the summation of each of the constitutive terms. System type n is $dM(0)/d\omega = -20n$

- Type 1:** $K_0(j\omega)^n$, $n = 0, \pm 1, \pm 2, \dots$
Magnitude: Draw a line with slope $= 20n$ and intercept $= 20\log(K_0)$
Phase: Draw a horizontal line at $\phi = n(90^\circ)$
- Type 2:** $(j\omega\tau + 1)^{\pm 1}$ $\omega_{BP} = 1/\tau$
Magnitude: Below ω_{BP} , the magnitude is 1 (0 dB).
Above ω_{BP} , behaves like type 1.
Phase:
 - for $\omega\tau \ll 1$, then $\angle(j\omega\tau + 1)^{\pm 1} = 0^\circ$
 - for $\omega\tau \gg 1$, then $\angle(j\omega\tau + 1)^{\pm 1} = 90^\circ$
 - for $\omega\tau \approx 1$, then $\angle(j\omega\tau + 1)^{\pm 1} \approx 45^\circ$
- Type 3:** $\left[(j\omega/\omega_n)^2 + 2\zeta(j\omega/\omega_n) + 1\right]^{\pm 1}$ $\omega_{BP} = \omega_n$
The resonance at the peak is determined by ζ .
Magnitude: Same as type 2, but with slope ± 40
Phase: S-curves from 0 to $\pm 180^\circ$ crossing ω_n at $\pm 90^\circ$

Gain Margin

Look at the phase plot of $C(s)G(s)$ and find ω_{GM} where phase $= -180^\circ$. The distance between the plot and 0dB at $M(\omega_{GM}) = GM$. $GM > 0$ is stable.

Phase Margin

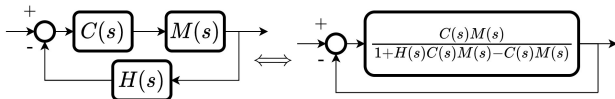
Look at the magnitude plot of $C(s)G(s)$ and find ω_{PM} where magnitude $= 0dB$. The distance between the plot and -180° at $\phi(\omega_{PM}) = PM$. $PM > 0$ is stable.

$$\text{if } T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\text{then } PM = \tan^{-1} \left(\frac{2\zeta}{[(1 + \zeta^4)^{1/2} - 2\zeta^2]^{1/2}} \right)$$

$$\text{if } 0 \leq PM \leq 60 \text{ then } PM = 100\zeta$$

To Unity



UNITY FEEDBACK ONLY

let $FP \triangleq$ forward path $\frac{E(s)}{R(s)} = \frac{1}{1 + FP}$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s)R(s) = \lim_{s \rightarrow 0} s \left[\frac{1}{1 + FP} \right] R(s)$$

Sys. Type	$R(s) = \frac{A}{s}$	$R(s) = \frac{A}{s^2}$	$R(s) = \frac{A}{s^3}$
0	$\frac{A}{1+K_P}$	∞	∞
1	0	$\frac{A}{K_v}$	∞
2	0	0	$\frac{A}{K_a}$
3	0	0	0

Match system type to input type to achieve $e_{ss} = 0$.

$$K_P = \lim_{s \rightarrow 0} FP \quad K_v = \lim_{s \rightarrow 0} sFP \quad K_a = \lim_{s \rightarrow 0} s^2FP$$

Analog Implementation

